

Reena Obmina

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Education

The University of Toronto

Expected June 2027

Bachelor of Science in Computer Science, Bioinformatics & Computational Biology

Toronto, ON

Awards: University of Toronto Scholar

Coursework: Software Design, Operating Systems, Theory of Computation, Computer Architecture & Organization, Probability, Statistics & Data Analysis, Software Tools & Systems Programming, Computer Graphics, Data Structures & Algorithms, Software Engineering, Bioinformatics, Molecular & Cell Biology

Technical Skills

Programming Languages: Python, Java, C, C++, Kotlin, R, SQL, HTML, CSS, JavaScript, MIPS Assembly

Libraries & Frameworks: Flask, SQLAlchemy, Spring Boot, PyTorch, pandas, NumPy, networkx, tidyverse, ggplot2, Plotly, Shiny, React, Jetpack Compose, Tailwind CSS, OpenGL, Matplotlib, JUnit

Development Tools: Git, GitHub, VS Code, RStudio, Android Studio, JetBrains Toolbox, Docker, Firebase, Linux (GCC, Make, Valgrind, gprof, SSH/SCP, CLI tools), Figma, Canva, Adobe Creative Cloud, Microsoft Office, Google Workspace

Databases: PostgreSQL, Firebase

Lab Techniques: PCR, Gel Electrophoresis, Micropipetting, Vegetation Surveys, DNA Extraction, Spectrophotometry, Aseptic Technique

Certifications: Good Clinical Practice (GCP), CPR-C + AED, TCPS 2: CORE, WHMIS

Research Experience

Student Researcher

September 2025 – Present

University of Toronto – Department of Cell & Systems Biology

Toronto, ON

Under the supervision of Dr. Nicholas Provart & Vincent Lau

- Contribute to bioinformatics research exploring plant gene expression and environmental stress responses within the Bio-Analytic Resource (BAR) platform
- Refactor legacy CGI and Perl scripts into modular Python Flask REST API, reducing average endpoint response times by over 98% (from 1019ms to 18ms) and improving scalability for multi-species genomics queries
- Develop automated XML validation scripts to detect malformed entries across 48+ API endpoints, identifying 14 critical failures and ensuring data integrity for over 30 plant species datasets
- Engineer performance benchmarking pipelines using ApacheBench and Python scripts to profile and optimize database queries, caching, and asynchronous request handling
- Author 20-page literature review analyzing FAIR principles, RESTful architecture, and quality control methodologies for bioinformatics infrastructure
- Collaborate with graduate students and developers to modernize backend systems, integrating Flask APIs, SQL databases, and data validation workflows with React-based visualization modules

Student Researcher

April 2025 – August 2025

University of Toronto – Department of Ecology & Evolutionary Biology

Toronto, ON

Under the supervision of Dr. Benjamin Gilbert & Grace Walker Mitchell

- Collected and analyzed biodiversity data across fragmented prairie habitats at the Koffler Scientific Reserve to study species dominance and extinction dynamics
- Applied statistical and visualization methods in R (tidyverse, ggplot2) to evaluate population fluctuations, demographic stochasticity, and community stability patterns
- Developed interactive data visualizations using R Shiny and Plotly to explore temporal and spatial patterns in species abundance across treatment groups
- Presented findings through research poster at Arts & Science Undergraduate Research Poster Fair with integrated QR codes linking to interactive graphs for dynamic data exploration

Presentations & Scholarly Work

Software Engineering Showcase

December 2025

- Presented *TurboType*, an AI-powered adaptive typing game at Department of Computer Science Showcase
- Collaborated with team of six students; delivered live demonstration showcasing intelligent sentence generation and adaptive difficulty using Clean Architecture principles under Professor David Jorjani

Literature Review & Project Proposal

October 2025

- “*Modernising ePlant: API Redevelopment, Quality Control, and Performance Optimisation for the Bio-Analytic Resource*”
- Authored 20-page scholarly review for BCB330 analyzing FAIR principles, RESTful architecture, and quality control methodologies for bioinformatics infrastructure modernization

Research Poster Presentation

September 2025

- “*Dominance Dynamics in Small Fragments: Demographic Stochasticity and Alternate Stable States in Restored Species*” presented at University of Toronto Arts & Science Undergraduate Research Poster Fair
- Co-authored with Julia Caccamo and Nanor Pontigian; designed comprehensive poster synthesizing field data, statistical modeling in R, and ecological theory with embedded QR code linking to interactive R Shiny dashboard

Projects

Bio-Analytic Resource (BAR): ePlant 3.0 | *Python, Flask, SQLAlchemy, React*

September 2025 – Present

- Modernize BAR’s genomics infrastructure by developing Python Flask APIs for querying multi-terabyte plant expression datasets
- Implement automated validation pipelines ensuring data integrity across 30+ species and 48+ API endpoints with standardized JSON schemas

TurboType | *Kotlin, Jetpack Compose, Firebase, Clean Architecture*

October 2025 – November 2025

- Co-engineered AI-powered adaptive typing platform with team of six students; implemented real-time difficulty scaling, multiplayer modes, user statistics, leaderboards, and personalization features
- Applied Clean Architecture principles with Firebase authentication and Jetpack Compose to deliver an engaging learning experience

Partial Word Writer | *Python, Flask, Spring Boot, Natural Language Processing*

September 2025 – October 2025

- Built cross-platform NLP-driven typing assistant for users with speech and motor barriers; implemented probabilistic text prediction reducing keystrokes by up to 55%
- Developed Flask-based APIs and assistive UI improving communication speed and accessibility

ML Epidemiology Engine | *Python, PyTorch, pandas, NumPy*

August 2025 – September 2025

- Designed machine learning workflow processing large epidemiological datasets to predict viral transmission patterns with high accuracy
- Created interactive dashboards enabling health and research teams to explore outbreak scenarios and transmission trends

Professional Experience

Volunteer

October 2025 – Present

Women’s College Hospital

Toronto, ON

- Support hospital operations and patient-centered initiatives; assist staff and visitors in navigating clinical and research spaces
- Engage with community health programs promoting accessibility, inclusion, and preventive care

Social Events Associate

July 2024 – Present

Computer Science Student Union – University of Toronto

Toronto, ON

- Plan and coordinate large-scale social and networking events for 100+ Computer Science students to strengthen community engagement
- Manage event logistics, budgeting, and communications; support promotional campaigns increasing student participation

Cashier

September 2022 – August 2023

Shoppers Drug Mart

Toronto, ON

- Delivered efficient customer service in fast-paced retail environment; operated POS systems and managed cash transactions with accuracy
- Assisted in inventory management and daily store operations; strengthened teamwork and communication skills